

Miracle Nnadiukwu

Software Engineer

+234 816 592 4062 | nnadiukwu@gmail.com | linkedin.com/in/miracle-nnadiukwu | github.com/dominioncruz

EDUCATION

Pan-Atlantic University - (First Class Honors)

Lagos, Nigeria

Bachelor of Science in Computer Science

Oct 2021 – Expected Dec 2025

Thesis: Occlusion-Sensitive Emotion Detection using GAN Augmentation

Apr 2025 - July 2025

- Proposed a novel occlusion-aware FER pipeline combining **PyTorch**, **OpenCV**, and **MediaPipe** for landmark detection, integrating **GAN inpainting at inference** (DeepFill v2) to restore occluded facial regions.
- Achieved a **10% accuracy increase** over baseline FER models under occlusion.

RESEARCH & PUBLICATIONS

Undergraduate Thesis | Department of Computer and Information Sciences

(2025)

- Investigated the effect of facial occlusion on FER systems and developed a GAN-augmented recognition pipeline.
- Conducted experiments on benchmark datasets, demonstrating the effectiveness of generative inpainting for occlusion recovery.
- Produced a reproducible CNN pipeline with potential applications in human-computer interaction and assistive technology.

Manuscript in Preparation | Occlusion-Sensitive Emotion Detection using GAN Augmentation

(2025)

- Drafting for submission to the Nigerian Journal of Technological Development (NJTD).
- Focus: evaluation of landmark-driven GAN pipelines for occlusion recovery in facial recognition and broader computer vision applications.

PROFESSIONAL EXPERIENCE

Full-Stack Developer | Beta Tenant NG

Aug 2025 - Present

- Building a PropTech platform addressing transparency and accessibility in Nigeria's real estate sector.
- Designing and implementing full-stack solutions with NestJS, PostgreSQL, and React.
- Applying research-driven design to features like listing verification and communication pipelines.

Back-end Engineer | Peak Harmony LTD

Aug 2024 - May 2025

- Researched and developed the company's core system architecture, implementing scalable back-end services with Node.js, PostgreSQL, and Redis.

Full-Stack Developer | Aku-Fintech

Aug 2023 - Oct 2023

- Designed and optimized back-end services with Node.js and hybrid databases, improving data retrieval efficiency by 15%.
- Strengthened system reliability through research-informed design improvements.

RECENT PROJECTS

ASL sign language detection | Pytorch

May 2025 - May 2025

- Built a deep learning model using PyTorch to recognize American Sign Language (ASL) gestures from image input, focusing on real-time detection accuracy.
- Trained a (CNN) from scratch and fine-tuned hyperparameters to achieve over 90% test accuracy.

PAU Pulse (Codespark hackathon) | NestJS, PostgreSQL, Gemini 2.0 Flash

Apr 2025 - May 2025

- Architected a scalable backend (NestJS, PostgreSQL) for an AI-driven news aggregator, reducing delivery latency by 14% for 100+ subscribers.

PAU Elearning | NestJS, PostgreSQL, Micro-services

Dec 2024- Jan 2025

- Incorporated Domain Driven Design as a system analyst to redesign school's elearning platform
- Increased fault tolerance and throughput using Micro-service architecture.

Decentralized voting system (TIC hackathon) | NestJS, MySQL, Web3.js, Ganache

Sept 2024 - Oct 2024

- Developed a voting system DApp on the Ethereum network aimed at ensuring secure, transparent elections.

Other notable projects

Varying

- Inventory management system (built with Java), Online Assessment System (built with Java), ProbHub(React Native, ExpressJS), PAU Internship site(full-stack app with NestJS and Vite), LL(1) Parser

COMMUNITY DEVELOPMENT

Mobile Development Lead | *Tech Innovation Club - Pan-Atlantic University*

May 2024 – Feb 2025

- Led a team of six members, teaching and building collaborative projects using React Native

Volunteer Administrator | *Elevate them program - Pan-Atlantic University*

Feb 2024 – Oct 2024

- Coordinated learning support for secondary school students in Ibeju-Lekki LGA, ensuring access to educational tools and resources provided by Pan-Atlantic University, fostering peer-to-peer learning and knowledge sharing.

Tutor | *Peer-To-Peer Tutorials - Pan-Atlantic University*

Mar 2023 – Mar 2024

- Organized and led weekly tutorials for undergraduates in core Computer Science subjects, including Advanced Mathematics, Core Programming Languages.
- Supported students in strengthening foundational skills, contributing to improved academic performance.

Additional work

Varying

- Admin (SIBAS team - Pan-Atlantic University), Volunteer (DevFest - Pan-Atlantic University)

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, Rust, Solidity, Typescript

Frameworks & Libraries: JavaFX, NestJS, Pytorch, ShadCN, Tailwind-CSS, React Native

Developer Tools: Camunda, Docker, Elastic stack, Git, Kafka, Kubernetes, Maven, Nginx, Remix

Databases: MongoDB, MySQL, PostgreSQL, Redis

Soft Skills: Adaptability, Collaboration, Communication, Leadership, Research, Time and Task Management